

RESEARCH ARTICLE

Enhanced Deep Learning-Based Optimization Model for the Coverage Optimization in Wireless Sensor Networks

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Abstract

A key component of wireless sensor networks (WSN) is optimal coverage. For WSN to have network lifetime and optimal resource utilization, maximum coverage must be guaranteed. The unequal distribution of sensor nodes in densely populated regions contributes to the build-up of network coverage. This research suggests a revolutionary intelligent deep learning-based optimization approach for WSN coverage optimization to tackle this problem. Here, the optimal network coverage of WSN is performed by the novel enhanced deep Q-network (EDQN) algorithm, where the parameters of DQN are tuned by nature inspired optimization algorithm called hippopotamus optimization (HO) with the intention of attaining the fitness function. In order to maximize coverage, the node position is changed in the developed methodology to represent the spatial properties of the network. Along with lowering latency, the increased coverage also increases throughput and network longevity. The outcome of several network coverage optimization trials is computed, and the EDQN-HO's impact on network coverage optimization is also determined by adjusting the parameters. The network coverage optimization studies' simulated findings demonstrate that the suggested EDQN-HO may be effectively utilized in a variety of settings. The proposed EDQN-HO for the WSN coverage model returns superior outcomes with 18.31%, 78.95%, 10.65%, 87.5%, 83.33%, and 38.20% than the existing methods in terms of coverage rate, energy consumption, computing time, positioning error, network lifetime, and average moving distance respectively.

Keywords Wireless sensor networks · Coverage optimization · Enhanced deep Q-learning · Hippopotamus optimization · Network lifetime

1 Introduction

Wireless sensor networks (WSNs) are a collection of spatially distributed sensor nodes designed to monitor and record environmental conditions, relaying data to a centralized base station. These networks are widely utilized in various domains, including environmental monitoring, healthcare, agriculture, and smart cities [1]. Coverage is a critical metric in WSNs, as it determines the extent to which the monitoring area is observed by the sensor nodes. Effective coverage not only ensures accurate data collection but also minimizes blind spots, enhances energy efficiency, and extends the network lifetime [2]. Achieving optimal coverage is inherently challenging due to resource constraints such as limited energy availability, bandwidth, and computational power in sensor



nodes [3, 6]. Moreover, the dynamic nature of real-world environments which is characterized by node failures, environmental changes, and uneven node distribution, further complicates the problem [4, 5].

A well-structured WSN involves careful placement of sensor nodes to ensure balanced coverage while addressing constraints like energy consumption and communication reliability [7, 8]. Advanced optimization techniques, including heuristic and machine learning-based methods, have been explored to tackle coverage issues [9, 10]. Neural networks (NNs) and Q-learning have shown promise in enhancing WSN coverage by enabling nodes to learn and adapt to dynamic conditions [11, 13, 15]. However, traditional solutions often fail to adapt to dynamic scenarios, leading to inefficient resource utilization and reduced network performance [12, 14].

In this work, we propose a novel enhanced deep Q-network (EDQN) algorithm optimized with the hippopotamus optimization (HO) technique to address these challenges. The proposed method dynamically adjusts node positions based on spatial and temporal properties of the network, achieving superior coverage, reduced energy consumption, and increased network longevity. The unique combination of deep learning and biologically inspired optimization offers a robust and scalable solution for WSN coverage optimization.

The paper contribution is as below.

- To suggest a revolutionary intelligent deep-learning-based optimization approach for WSN coverage optimization.
- To perform the optimal network coverage of WSN by the novel EDQN, where the parameters of DQN are tuned by nature inspired optimization algorithm called HO with the intention of attaining the fitness function.
- To change the node position for maximizing coverage in the developed methodology to represent the spatial properties of the network.
- To increase throughput and network longevity along with minimal latency and increased coverage.

The paper organization is as follows. Section 1 is the introduction of WSN coverage model. Section 2 is a literature survey. Section 3 is proposed methodology with proposed model, network model, objective function, WSN coverage model, proposed EDQN algorithm, and HO algorithm. Section 4 is the results. Section 5 is the conclusion.

1.1 Motivation

WSNs play a key role in many areas such as disaster response urban planning and tracking the environment. However, power limits, sensor placement, and changing conditions still make it hard to get good coverage in WSNs. Regular coverage methods often cannot handle these complex issues leading to wasted resources and poor network performance. One promising way to boost WSN coverage involves using deep learning-based optimization algorithms. By tapping into deep learning's strengths, we can create systems that learn from big data sets to predict the best sensor setups and spots on the fly. This approach allows coverage strategies to adapt in real time to changes in the network and surroundings. In addition, deep learning can help manage power use well making sensor nodes last longer while keeping good coverage. This research aims to change how we run WSNs and enable tougher more effective systems that can meet shifting needs. The main goal is to build smart WSNs that work at their best and meet coverage needs paving the way for smarter and more responsive tech solutions.

2 Related Work

This study looked into the connected target coverage (CTC) problem with a focus on making the best use of energy [16]. The researchers came up with a new two-step way to optimize, which made sure users stayed connected and targets were covered for as long as the network lasted. In the first step of their approach, they used a new multi-goal integer linear programming (ILP) model to arrange sensors into as many separate cover sets (CSs) as

possible. A subset of possible cluster heads (CHs), or the group of remaining nodes that were unable to form an independent CS, were assigned to the current CSs. In other words, CHs were selected from the potential CH subset for each active CS. To collect data from targets, the algorithm initiated CSs one after another in the second stage.

To maximize the efficient coverage area with acceptable BER, the suggested BER-conscious array synthesis approach took predicted power as well as SIR restrictions into account throughout the antenna design procedure [17]. Reliable connectivity and optimized coverage implied energy savings at the sensor nodes. An analytical synthesis was performed on a multi-sector AP antenna on the basis of the suggested coverage optimization approach. The analytically calculated coverage values were validated by experimentation and a prototype was constructed. Additionally, the communication region's high port isolation of 45.95 dB within neighboring sectors allowed for the creation of numerous simultaneous beams, which boosted system capacity. The synthesized switched multi-beam multi-sector antenna utilizing the suggested technique was a promising option for AP in WSN applications, as evidenced by the achieved effectiveness.

To increase data transmission reliability and energy efficiency, research focused on developing and refining energy-efficient coverage strategies for WSN nodes [18]. The manuscript examined how to create an energy-efficient sensor network node coverage method by thoroughly investigating and analyzing flat and hierarchical routing protocols. This involves ensuring that each monitoring point is covered by at least one sensor node. Simultaneously, the study investigated an energy-efficient coverage technique on the basis of the refined grey wolf algorithm with the goal of optimizing sensor node placement and boosting node coverage efficacy. Studying efficient methods aimed at WSN's energy consumption would help to enhance the resource utilization efficiency and the relative satisfaction of the demand for intelligent monitoring and control in the future.

In this study, a probabilistic model-based 3D coverage deployment method based on the received signal strength in the probabilistic model was suggested [19]. The distance between the nodes might be generally computed from the routing loss presumably linked with the wireless signal during the propagation process. The maximum distance within the nodes was defined through introducing a threshold with regards to the path loss, providing robustness and reliability of the network additionally. The probability coverage was used in addition to build the cube-oriented network coverage. Based on this, the problem of optimum coverage in 3D-WSN was studied. The traditional particle swarm optimization could be optimized in such a way that it combined the two methods and reached the global optimum much faster while avoiding local optimum convergence. The simulation results proved, accordingly, as to how quickly the highlighted strategy might converge towards enhancing network coverage across the network.

Interestingly, this work enhanced the DV-Hop localization method using an improved sparrow search pattern [20]. In the research, the less mean square error (MSE) criteria was used to reduce errors in computation and the researchers added a deviation correction factor with the view of changing the minimum number of node hops. Further, the research helped to improve the sparrow search algorithm by the initiation of GPS optimization population. Again, GSSA emerged top in the algorithm performance comparison and was distinguished by the optimal convergence effectiveness. The research presented a novel method of DV-Hop optimization along with a benchmark model to use DV-hops for WSN coverage.

To achieve better area coverage, an effective approach was defined which might use a few static nodes along with a number of mobile nodes [21]. Achieving wide area coverage necessitated strategic placement of the static nodes, and channeling of the mobile nodes in a specific direction. To achieve this, three models of mixed integer linear programming (MILP) were presented below. Whereas the other two formulations defined how to steer a set of mobile nodes to achieve the maximum spatial coverage and minimum number of steps required to achieve the specific coverage, the first formulation essentially used a set of static nodes. The efficiency analysis of the suggested algorithms was given indicating how it differs from the typical parameters in the industry.

Based on the Reptile Search Algorithm RSA, it was possible in this paper to propose a new approach to optimizing WSN coverage [22]. The Reptile Search method could benefit several procedures as it has been implemented on enhancing issues in the past. Bio-economic properties of non-optimal people were disregarded by the RSA, which has to track the path of ideal person in each iteration. To fully mine all of the positional

data buried in the whole population, the article incorporated a distribution estimation approach into the RSA methodology. To assess the viability of the suggested approach, a number of functions were chosen to serve as benchmark functions for optimization tests. In this study, certain conventional optimization techniques including the standard RSA were compared with the suggested enhanced RSA.

A novel ruminant algorithm, inspired from biological features, was presented in this research work [23]. The efficient digestive systems than the monogastric animals were capable of informed the creation of the algorithm. Organic food production fashioned an excellent food value that was rich in energy from the massive amount of raw food as input. The proposed algorithm has been tested under many reference functions as the results indicate. Thus, the suggested method of maximizing the distribution of sensor and power nodes to increase the coverage of WSNs and prolong the performance of equipment work, was done with the motivational factor in mind. In addition, based on the algorithm, it was proved that even if no more sensors are deployed in the field, the optimized energy value enhanced with the extension of network coverage.

For coverage optimization with an energy efficient approach, the multi-strategy grey wolf optimization (MSGWO) algorithm was used [24]. This technique might further reduce energy consumption and enlarge coverage area if the sort-driven hybrid opposition-oriented learning proceeded in parallel with the higher-order multinomial sensing methods. In addition, boundary and node movement algorithms were discussed to help nodes synch their movement with those installations that were obstacle-aware if the latter encountered barriers. Message was gradually gotten into the form as efficient as predicted for the MSGWO, which was further validated and compared based on numerous traditional test functions.

To improve the coverage issue and especially in 2-D and more complex 3-D zones, an adaptive learning grey wolf optimizer (ALGWO) was introduced [25]. To avoid the problem of premature convergence and improve the exploration capability of the algorithm, ALGWO utilized a dynamic and space variant search strategy together with the dynamic reverse learning approach. In providing people the knowledge of the Northeast dimension, adaptive dimensional learning successfully freed people from the constraints of the first three wolves and diversified the population. For the meantime, all the participants' dimensional learning applied exploitation and exploration in a flexible manner. Still, specifically, the ALGWO performance was checked in the 2D and 3D WSN environment with the help of simulations. According to the outcomes of the simulation, the ALGWO algorithm demonstrated better characteristics in terms of coverage than GWO as well as its improved versions. Load balancing and possible solutions have also been extensively discussed in the literature [26, 27].

2.1 Problem Statement

Providing proper coverage in WSNs is definitely stating a major concern which is needed for data gathering and monitoring in a variety of applications. Traditional solutions often entail employing pre-determined algorithms, which are not compatible with dynamic scenarios or node distribution changes—they result in uncovered areas and inefficient power consumption. Since most sensor nodes are constrained in terms of resources, there is need to come up with strategies that enhance coverage and, at the same time, reduce energy consumption. The existing coverage approaches fail to consider the dynamics of real-world conditions such as node failure, change in topography, and variations in the physical environment. For this reason, the network has lesser efficiency and can only have a limited lifetime. Furthermore, these traditional approaches to optimization are not able to adjust their methods to accommodate these new problems in the same way and do not possess the flexibility necessary for future adjustment. The purpose of this work is to present an optimal approach to coverage of WSN using deep learning that will consider these constraints. This needs to be done with the aim of coming up with a flexible system that is capable of predicting probable efficient locations for sensors, using the previous data to make future coverage strategies, and incorporating deep learning techniques. This strategy will assist in helping to enhance energy management together with coverage effectiveness, which will in turn be useful in the development of more intelligent as well as more reliable WSNs.

3 Proposed Methodology

3.1 Proposed Model

The proposed model will strengthen and optimize the utilization of various deep learning techniques for sensor coverage in relation to WSNs in consideration to the conceived and realized coverage notions and definitions. Accomplishing the intended level of coverage with the use of WSNs is vital for the proper assessment and gathering of the needed information. This model employs deep learning techniques for the optimization of the sensor coverage problem. In this way, a larger coverage area is reached with less energy. The model employs deep learning in conjunction with the optimization to achieve different terrains or the deployment of networks. It also enhances how these systems perform over time by overcoming issues like the failure of nodes or the depletion of low energy batteries. While the enhanced model of WSN improves coverage density, it also increases the lifetime of the network by looking for better resource distribution. Here, the unique EDQN algorithm achieves the best network coverage for WSN. To achieve the fitness function, the parameters of DQN are adjusted using the nature-inspired optimization method HO. The established approach modifies the node location to represent the geographical features of the network to maximize coverage. The enhanced coverage not only reduces latency but also boosts throughput and extends the life of the network. The results of several network coverage optimization experiments are calculated, and by varying the parameters, the effect of the EDQN-HO on network coverage optimization is also ascertained. This technique is a breakthrough in the management of WSNs allowing the implementation of advanced and more efficient monitoring systems in a variety of areas including environmental protection, smart cities, and industrial automation. The proposed WSN coverage model is shown in Fig. 1.

3.2 Network Model

In the monitoring area, n sensors are assumed to be assigned and complete sensor nodes form a group $T = \{o_1, o_2, \dots, o_j, \dots, o_{n-1}, o_n\}$, in which o_j indicates the j^{th} node with its coordinate as (y_j, z_j) . The likelihood that node o_j will find the target point u for every point q with coordinates (y_q, z_q) in the monitoring zone may be represented as

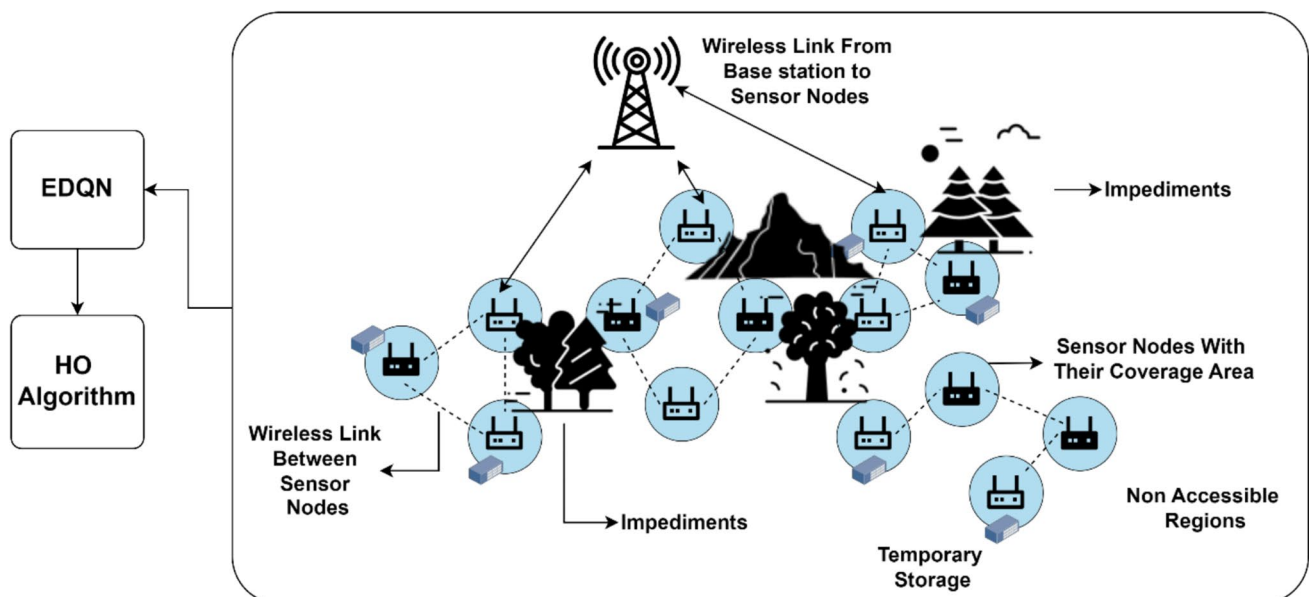


Fig. 1 Proposed WSN coverage model

$$Q_u(o_j, u) = \begin{cases} 1 & e(o_j, u) \leq s_t - s_f \\ \exp\left(\frac{-\lambda_1 \alpha_1^{\beta_1}}{\alpha_2^{\beta_2} + \lambda_2}\right) & s_t - s_f < e(o_j, u) < s_t + s_f \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Here, s_f represents the sensing reliability parameter of the single sensor node, $0 < s_f < s_t$, $\alpha_1 = s_f - s_t + e(o_j, u)$, $\alpha_2 = s_f - s_t - e(o_j, u)$, λ_1 , λ_2 , β_1 , and β_2 are all sensing coefficients associated with node characteristics, s_t represents the sensing range, and $e(o_j, u)$ represents the Euclidean distance between node o_j and the target point u . The total likelihood that every sensor node in the monitoring area will identify the target point u is

$$Q_u(o_{All}, u) = 1 - \prod_{j=1}^n (1 - Q_u(o_j, u)) \quad (2)$$

Here, o_{All} represents the collection of entire sensors that are able to identify point u as the target. The monitoring region is divided into $m \times o$ grids in order to calculate the coverage rate for the whole network. Every grid is shaped like a rectangle section, with every section having an equal size of $1n \times 1n$ and being simplified to a pixel having discretized precision of $1n^2$. Therefore, the ratio related to the detectable grid point count to the total grid point count defines the coverage rate associated with the WSN:

$$D = \frac{\sum_{y_u=1}^m \sum_{z_u=1}^o Q_u(o_{All}, u)}{m \times o} \quad (3)$$

Another metric for assessing WSN effectiveness is evenness. A more uniform node distribution, which results in reduced energy use and a longer lifetime, is shown by more evenness. The single node o_j 's evenness is calculated as

$$V_j = \sqrt{\frac{1}{l_j} \sum_{k=1}^{l_j} (E_{j,k} - N_j)^2} \quad (4)$$

Here, N_j represents the average of all the distances between node o_j and all of its neighbor nodes, $E_{j,k}$ represents the distance from node o_j to node o_k , and l_j represents the count of neighbor nodes. The WSN's overall evenness is

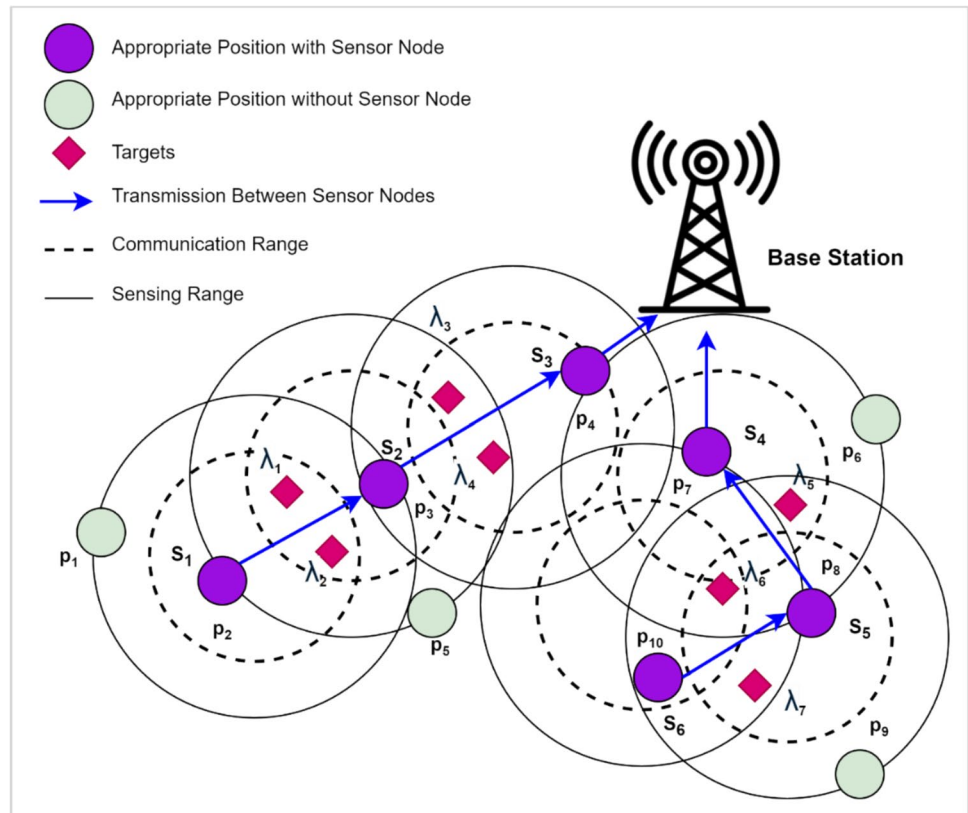
$$V = \frac{1}{n} \sum_{j=1}^n V_j = \frac{1}{n} \sum_{j=1}^n \sqrt{\frac{1}{l_j} \sum_{k=1}^{l_j} (E_{j,k} - N_j)^2} \quad (5)$$

Here, n represents the total count of nodes. Equation 5's formulation states that a lower V value corresponds to a higher evenness. Additionally, the average moving distance—which is described as—is utilized to assess the overall energy consumption of entire moving nodes.

$$E_n = \frac{1}{n} \sum_{j=1}^n \sqrt{(y_j - y_0)^2 + (z_j - z_0)^2} \quad (6)$$

Here, y_j, z_j represent the updated values of node o_j following coverage optimization, and y_0, z_0 represent the starting coordinates related to node o_j before to coverage optimization. The diagrammatic model of target-oriented WSN with coverage is shown in Fig. 2.

Fig. 2 Target-oriented WSN with coverage



3.3 Objective Function

The objective function is a prime factor of the proposed research, as it translates the challenges of WSN coverage optimization into actionable goals. It addresses three critical aspects: minimizing energy consumption to extend network lifetime, maximizing coverage ratio to reduce blind spots, and ensuring energy balance across nodes to maintain consistent performance. By balancing these conflicting objectives, the function guides the optimization process effectively, enabling the EDQN-HO algorithm to achieve superior results in terms of coverage efficiency, energy usage, and network longevity. This comprehensive and adaptable formulation ensures the proposed methodology is robust and practical for real-world WSN applications.

The subject of WSN coverage optimization has been developed as a multi-objective optimization problem (MOP). It is distinguished by the existence of many objectives that could be at odds with one another. Improving the effectiveness of one MOP aim might perhaps result in a decline in effectiveness for one or multiple objectives. The MOP problem, which has to do with a WSN's sensor node scheduling, represents the main subject of this investigation. The goal is to optimize energy usage and guarantee a balanced distribution of energy throughout the entire network by carefully selecting active nodes, CH nodes, and idle nodes to maximize the coverage of target areas.

Considering that g represents a node scheduling solution, three goals are formulated for the MOP issue. Energy consumption, coverage ratio, and energy balance are all included in these objectives. Two main ideas are included in the notion of energy equilibrium: regional energy and energy balance. The observation area is divided into A zones, represented by the symbol $a \in [1, 2, \dots, A]$. Equation (7) states that the mean residual energy of entire nodes within the zone equals the regional energy in the a^{th} zone FR_a .

$$FR_a = \frac{\sum_{j=1}^{o_a} F_{a_j}}{o_a} \quad (7)$$

Here, F_{a_j} is the energy left in the j^{th} node of the a^{th} grid, and o_a is the total count of nodes in the a^{th} grid. The energy balance measure for solution T is denoted by the sign $Ft(T)$ in the area of the investigation. The percentage of energy difference among the network's top as well as lowest regional energy levels in relation to the highest regional energy level is measured by this statistic. A fairer distribution of energy consumption within the network is indicated by a lower value of $Ft(T)$. Mathematically, Eq. (8) defines $Ft(T)$:

$$Ft(T) = \frac{\max(FR_a) - \min(FR_a)}{\max(FR_a)} \quad (8)$$

The energy level linked with region a is denoted by FR_a , whereas the greatest energy level across entire regions is represented by $\max(FR_a)$ and the minimum energy level across entire regions is represented by $\min(FR_a)$. In WSN, maintaining a high degree of coverage rate is crucial. The goal of this study is to change the original goal of raising the coverage percentage into the job of decreasing the number of desirable spots $O(M)$ that are left uncovered, as given by Eq. (9).

$$O(M) = \sum_{u \in U} V(u^{(t)}) \quad (9)$$

$$V(u) = \begin{cases} 0 & \text{if } \exists t_j \in T_{Active} \text{ and } (t_j, u) \leq s_j \\ 1 & \text{else} \end{cases} \quad (10)$$

Equation 10 states that the variable $V(u)$ is used to determine the target point u 's coverage status. The number of target points is represented by the variable U , while the group of active sensor nodes is indicated by the variable T_{Active} . In the framework of solution T , energy consumption represents the overall amount of energy used for operations like signal aggregation, signal reception, signal transmission, and sensor activation. This may be quantitatively stated using the formula in Eq. 11.

$$F(g) = \left(\sum_{j=1}^{Num} \sum_{T \in D_j} F_{T, ch_j} + F_{SY} + F_{EB} \right) + \sum_{j=1}^{Num} F_{ch_j, sink} + F_{Total} \quad (11)$$

Here, Num is the count of clusters and D_j is the j^{th} cluster. F_{T_j, T_k} is defined by Eq. (12) as the energy used by node T_j to transfer data to node T_k .

$$F_{T_j, T_k} = \begin{cases} m \times F_{Elec} + m \times F_{gt} \times e^2, & e < e_0 \\ m \times F_{Elec} + m \times F_{nq} \times e^4, & e \geq e_0 \end{cases} \quad (12)$$

The energy used by the transceiver circuitry is represented by the letter F_{Elec} . Free space and multipath fading methods define F_{gt} and F_{nq} as the energy used during 1-bit data transmission in order to maintain a reasonable Bit Error Rate (BER), respectively. The multipath method is used in other situations and the free space method is used when the distance within two nodes is less than the threshold distance, e_0 . Equation (13) specifies the calculation of both F_{SY} and F_{EB} , which represent the energy consumed during the data receipt procedure and data aggregation, respectively.

$$F_{SY} = F_{EB} = F_{Elec} \times m \quad (13)$$

The entire amount of energy used to activate every node in the current round is called F_{Total} , and it is calculated using Eq. (14).

$$F_{Total} = \sum_{j=1}^o F_{BD} \times b_j \quad (14)$$

The energy required to turn on an inactive node is represented by F_{BD} . Equation (15), which specifies the conditions under which the variable b_j is to be determined, controls the indication of whether or not the node j is in an active state.

$$b_j = \begin{cases} 1 & \text{if } T_j \in T_{Nonch} \text{ or } T_j \in T_{ch} \\ 0 & \text{else} \end{cases} \quad (15)$$

3.4 WSN Coverage Model

Assuming that the monitoring region is a 2D target plane, O sensor nodes were placed within it, each having a detecting radius of s and a transmission radius of S . For network connectivity, the transmission radius is typically placed at $S = 2s$. Formula (16) may be used to represent the group of sensor nodes:

$$H = \{h_1, h_2, h_3, \dots, h_o\} \quad (16)$$

Here, the coordinates associated with every node are (b_j, c_j) . The monitoring zone can be divided by using $N \times o$ pixel points. The distances among G and every sensor node h_j were determined using formula (11) and the pixel point G coordinates, which were y_G, z_G .

$$e(h_j, G) = \sqrt{(y_G - b_j)^2 + (z_G - c_j)^2} \quad (17)$$

Here, the distance between sensor node h_j and point G is indicated by $e(h_j, G)$. The probability of being seen by the sensor node h_j at pixel position G (18) is calculated using the following formula:

$$q(h_j, G) = \begin{cases} 1 & e(h_j, G) \leq s \\ 0 & e(h_j, G) > s \end{cases} \quad (18)$$

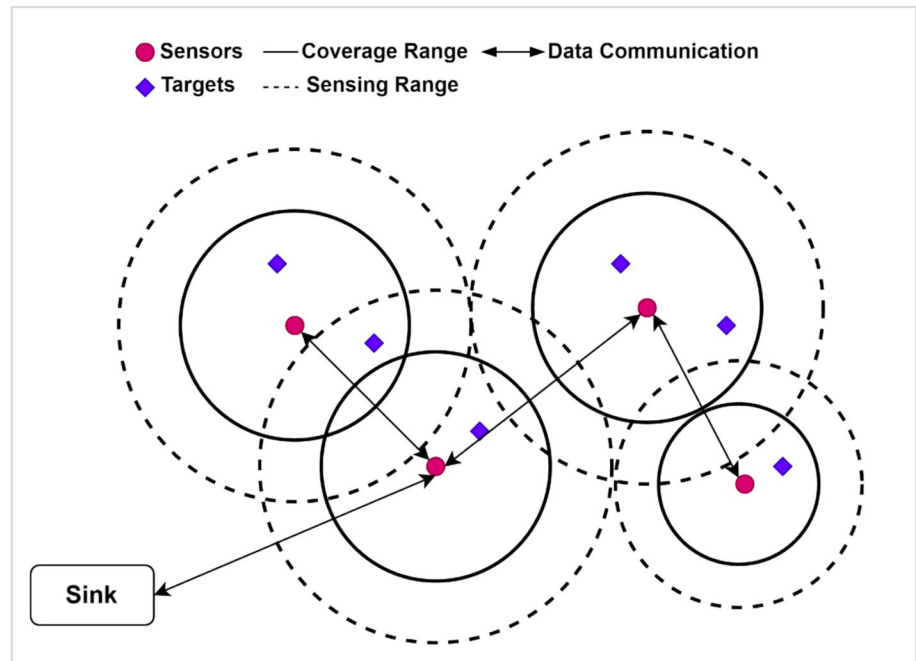
Here, $q(h_j, G)$ represents the probability of being picked up by the sensor node h_j at pixel location G , and s represents the sensing radius. Formula (19) shows the joint likelihood that a pixel G is received by a group of nodes H . In real-world circumstances, every pixel may be experienced by multiple sensor nodes.

$$q(H, G) = 1 - \prod_{h_j \in H} [(1 - q(h_j, G))] \quad (19)$$

The coverage ratio associated with the sensor node positioned in the monitoring region is shown by formula (20):

$$cov(H) = \frac{\sum_{G \in n \times o} q(H, G)}{n \times o} \quad (20)$$

Here, $q(H, G)$ represents the likelihood that the node group H will detect the pixel point G and $n \times o$ represents the total count of pixels. The WSN coverage model is diagrammatically shown as in Fig. 3.

Fig. 3 WSN Coverage model

3.5 Proposed EDQN Algorithm

The EDQN algorithm is selected here in the proposed WSN coverage model for performing the optimal network coverage, in which the parameters of DQN are optimized by HO with the consideration of deriving the objective function. One input layer, one output layer, and several hidden layers make up the DQN, a forward neural network (NN). The DQN algorithm selects and evaluates actions based on the same action reward value. It is necessary to assess the predicted value of function $R(t, b)$ for each action b taken for each state t in an episode. The Bellman optimality equation describes this function as follows:

$$R(t, b) = F \left(s + \gamma Q(t, t') \underbrace{\max_{b'}}_{b'} R(t', b') \right) \quad (21)$$

Here, b' represents the action in state t' , $Q(t, t')$ represents the probability of arriving at state t' from state t , and $F(\bullet)$ represents the expected value. The action reward values of every state-action combination are kept in a table and are continually updated iteratively using the conventional Q-learning process. When the state space is very vast, it becomes challenging to store and iterate over entire values of the state-action pairings. The DQN algorithm—which combines the Q-learning and deep learning algorithms—solves this issue. The DQN technique uses neural networks (NNs) to learn how to fit a parameterized approximation value function $R(t, b|\theta)$, in which θ represents the parameter of $R(t, b|\theta)$, to the true reward function $R(t, b)$.

The count of inputs is determined by the size associated with state space elements O_U , and its input represents the current state $t_u (t_u \in T)$ at time u . The estimated action reward values of functions $R(t_u, b|\theta) (b \in B)$ in that state t_u are all contained in its output, which has the same dimension as action space O_B . The action in the output layer with the highest reward value is chosen by the agent. In DQN, the loss function is written as

$$M_{DQN}(\theta_j) = F \left(z_j^{DQN} - R(t, b|\theta_j) \right)^2 \quad (22)$$

Here, the parameter associated with $R(t, b|\theta)$ at iteration j is denoted by θ_j , and the objective function is represented by z_j^{DQN} . The definition of the target function is:

$$z_j^{DQN} = s_j + \gamma R(t', b_{DQN}^* | \theta_j^-) \quad (23)$$

Here, as shown in (24), b_{DQN}^* represents the action in the DQN algorithm with the maximum approximate value in the state t' , s_j represents the reward at iteration j , θ_j^- represents the parameter, and $R(t', b_{DQN}^* | \theta_j^-)$ represents the maximum value in the state t' .

$$b_{DQN}^* = \underbrace{\arg \max}_{b'} R(t, b | \theta_j^-) \quad (24)$$

DQN amplifies WSN coverings patterns through real-time adaptive decision-making. It learns from environmental dynamics to optimize sensor placement and operation thereby improving coverage efficiency, reducing energy consumption, and prolonging network lifetime. In addition, DQNs are suitable for handling complex and high-dimensional state spaces which results in more robust WSN performance. Nevertheless, DQN encounters high computation complexity; high resource demands making them unfit for resource constrained nodes in WSN coverage models. Furthermore, DQNs need large amounts of training data and time which results in potential convergence problems leading to instability during dynamic environments hence complicating real-time decisions making processes. Therefore, to overcome the shortcomings, the parameters of DQN are tuned by HO with the intention of attaining the fitness function, thus known to be novel EDQN. This EDQN improve WSN coverages using advanced techniques such as experience replay and target networks leading to stable learning that effectively deals with non-stationary environments, optimizes resource allocation, and enhances scalability thus improving coverage efficiency, reducing energy consumption while improving overall network performance. The proposed EDQN algorithm for the WSN coverage model is shown in Fig. 4.

3.6 HO Algorithm

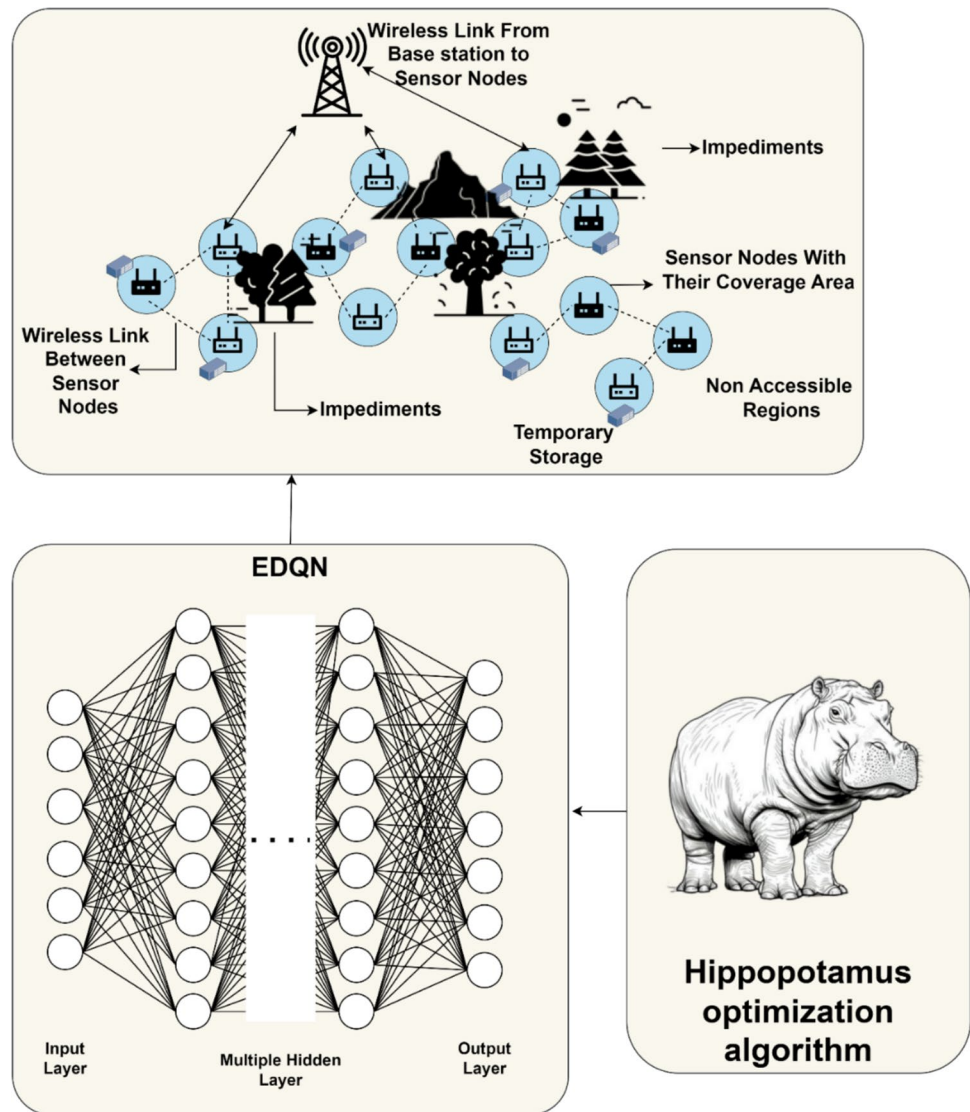
The HO algorithm is selected here in the proposed coverage optimization in WSN model to tune the DQN parameters with the consideration of attaining the optimal network coverage. The HO describes an inventive use of metaheuristic technique, since it was inspired by the innate behaviors of hippopotamuses. The HO is theoretically specified by means of a trinary-stage method that includes their mathematically formulated escape techniques, defense actions against predators, and location updates in ponds or rivers.

A population-oriented optimization technique called the HO utilizes hippopotamuses as search agents. Hippos are potential solutions for the optimization issue in the HO method, which means that every hippopotamus' position update in the search space reflects values for the decision variables. As a result, every hippo is described as a vector, and a matrix is used to mathematically describe the hippo population. The HO's initialization step includes creating randomized initial solutions, just as other optimization algorithms. In this stage, the below formula is used to construct the vector associated with decision variables:

$$Y_j : y_{jk} = LB_k + s \cdot (UB_k - LB_k), j = 1, 2, \dots, O, k = 1, 2, \dots, n \quad (25)$$

Here, s represents a random number between 0 and 1, Y_j represents the location associated with the j^{th} candidate solution, and UB and LB stand for the k^{th} decision variable's upper as well as lower bounds, respectively. Equation (26) forms the population matrix provided that O represents the count of hippopotamuses in the herd and n represents the count of decision variables in the issue.

Fig. 4 Proposed EDQN algorithm for the WSN coverage model



$$Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_j \\ \vdots \\ Y_O \end{bmatrix}_{O \times n} = \begin{bmatrix} y_{1,1} & \cdots & y_{1,k} & \cdots & y_{1,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ y_{j,1} & \cdots & y_{j,k} & \cdots & y_{j,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ y_{O,1} & \cdots & y_{O,k} & \cdots & y_{O,n} \end{bmatrix}_{O \times n} \quad (26)$$

Numerous adult female hippopotamuses, calves, numerous adult male hippopotamuses, and dominant male hippopotamuses (the herd leader) make up a hippopotamus herd. On the basis of the fitness function value iteration—the lowest for the minimization issue and the greatest for the maximization task—the dominating hippopotamus is identified. Hippocampal species typically congregate next to one another. Male dominant hippopotamuses defend their area and herd against intruders. The male hippopotamuses are surrounded by a number of female counterparts. Male hippopotamuses are expelled from the herd by the dominant male once they reach adulthood. Consequently, to establish their own dominance, these banished male individuals must either recruit females or compete with other established male members of the herd. The location related to the male hippopotamus herd members in the pond or lake is expressed mathematically in Eq. (27).

$$Y_j^{NHippo} : y_{jk}^{NHippo} = y_{jk} + z_1 \cdot (E\ Hippo - J_1 y_{jk}) \text{ for } j = 1, 2, \dots, \left\lceil \frac{O}{2} \right\rceil \text{ and } k = 1, 2, \dots, n \quad (27)$$

The male hippo's location is represented by Y_j^{NHippo} in Eq. (27), while the dominant hippo's location is shown by $E\ Hippo$ (the hippos with the highest cost in the present iteration). s_5 represents a random number between 0 and 1 (Eq. 28), J_1 and J_2 are integers between 1 and 2, and $\vec{s}_{1,\dots,4}$ represents a random vector between 0 and 1 (Eqs. 27 and 30). In Eq. (27), z_1 represents a random integer between 0 and 1, and nH_j represents the mean values of certain randomly chosen hippos having an equal likelihood of containing the present hippos under consideration (Y_j). ρ_1 and ρ_2 , which are random integer integers in Eq. (28), can have values of one or zero.

$$i = \begin{cases} J_2 \times \vec{s}_1 + (\sim \rho_1) \\ 2 \times \vec{s}_2 - 1 \\ \vec{s}_3 \\ J_1 \times \vec{s}_4 + (\sim \rho_2) \\ s_5 \end{cases} \quad (28)$$

$$U = \exp\left(-\frac{u}{U}\right) \quad (29)$$

$$Y_j^{GCHippo} : y_{jk}^{GCHippo} = \begin{cases} y_{jk} + i_1 \cdot (E\ Hippo - J_2 nH_j) & U > 0.6 \\ \Xi & \text{otherwise} \end{cases} \quad (30)$$

$$\Xi = \begin{cases} y_{jk} + i_2 \cdot (nH_j - E\ Hippo) & s_6 > 0.5 \\ LB_k + s_7 \cdot (UB_k - LB_k) & \text{otherwise} \end{cases} \text{ for } j = 1, 2, \dots, \left\lceil \frac{O}{2} \right\rceil \text{ and } k = 1, 2, \dots, n \quad (31)$$

Equations (30) and (31) delineate the location of female or immature hippos $Y_j^{GCHippo}$ in relation to the herd. The majority of young hippopotamuses are found close to their moms, but occasionally, out of curiosity, they stray from the herd or their mothers. U higher than 0.6 indicates that the young hippopotamus has separated from its mother (Eq. 29). Equation (31) states that if s_6 , a value between 0 and 1, is more than 0.5, the young hippopotamus has split from its mother but is still in or close to the herd. If not, it has completely separated. Equations (30) and (31) are used to simulate the behavior of young, female hippos. In the i equation, i_1 and i_2 represents random integers or vectors chosen from the five situations. A random number within zero and one is denoted by s_7 in Eq. (31). Equations (32) and (33) give the position update of the immature or male and female hippos within the herd. G_j represents the value of the fitness function.

$$Y_j = \begin{cases} Y_j^{NHippo} & G_j^{NHippo} < G_j \\ Y_j & \text{otherwise} \end{cases} \quad (32)$$

$$Y_j = \begin{cases} Y_j^{GCHippo} & G_j^{GCHippo} < G_j \\ Y_j & \text{otherwise} \end{cases} \quad (33)$$

The J_1 and J_2 situations in the suggested technique boost exploration as well as global search by utilizing i vectors. It improves the suggested algorithm's exploration procedure and results in a better global search.

Hippopotamuses' main defense maneuver is quickly turning to face the predator and making loud noises to scare it away from getting too nearby. Hippopotamuses may act in this stage by approaching the predator

and causing it to retreat, therefore, warding off any possible threat. The location of the predator in the search space is shown by Eq. (34).

$$predator : predator_k = LB_k + \vec{s}_g \cdot (UB_k - LB_k), k = 1, 2, \dots, n \quad (34)$$

Here, a random vector with a range of zero to one is represented by $\vec{s}_g$.

$$\vec{E} = |predator_k - y_{jk}| \quad (35)$$

Equation (35) shows how far away the predator is from the j^{th} hippopotamus. To defend itself from the predator during this period, the hippopotamus develops a protective behavior on the basis of the factor $G_{predator_k}$. The hippopotamus quickly turns and advances in the direction of the predator to force it to withdraw if $G_{predator_k}$ is less than G_j , which indicates that the predator is quite near to the hippopotamus. When $G_{predator_k}$ is higher, it means that the hippopotamus' domain is closer to the predator or other intruding object as in Eq. (36). The hippopotamus in this instance turns to face the predator, but with a smaller range of motion. The idea is to draw attention to its existence inside its region and alert any predators or intruders.

$$Y_j^{hippoS} : y_{jk}^{hippoS} = \begin{cases} \overline{SM} \oplus predator_k + \left(\frac{g}{(\sigma \times \cos(2\pi h))} \right) \cdot \left(\frac{1}{\vec{E}} \right) G_{predator_k} G_j \\ \overline{SM} \oplus predator_k + \left(\frac{g}{(\sigma \times \cos(2\pi h))} \right) \cdot \left(\frac{1}{2 \times \vec{E} + \vec{s}_g} \right) G_{predator_k} \geq G_j \end{cases} \text{ for } j = \left[\frac{O}{2} \right] + 1, \left[\frac{O}{2} \right] + 2, \dots, O \text{ and } k = 1, 2, \dots, n \quad (36)$$

A hippopotamus at the location of Y_j^{hippoS} faces a predator. $\overline{SM}$ describes a Levy-distributed random vector that is employed when there are abrupt shifts in the predator's location during a hippopotamus attack. Equation (37), which calculates the mathematical method for the random movement of Levy movement, is used. The variables x and w represent random integers in the interval $[1]$, respectively; ϑ represents a constant ($\vartheta = 1.5$), Γ stands for the Gamma function, and σ_x may be calculated using Eq. (38).

$$levy(\vartheta) = 0.05 \times \frac{x \times \sigma_x}{|w|^{\frac{1}{\vartheta}}} \quad (37)$$

$$\sigma_x = \left[\frac{\Gamma(1 + \vartheta) \sin\left(\frac{\pi\vartheta}{2}\right)}{\Gamma\left(\frac{(1+\vartheta)}{2}\right) 2^{\frac{(\vartheta-1)}{2}}} \right]^{\frac{1}{\vartheta}} \quad (38)$$

The uniform random numbers g and d in Eq. (36) are between 2 and 4, respectively, and between 1 and 1. E represents a consistent random number between 2 and 3. A uniform random number between -1 and 1 is represented by h . A random vector of size $1 \times n$ is represented as $\vec{s}_g$.

Equation (39) states that if G_j^{hippoS} is more than G , the hippo has been hunted and will be replaced in the herd by another hippo; if F is less than F , the hunter will flee, and the hippo will rejoin the herd. During the second stage, notable improvements were seen in the global search procedure. The first as well as second stages work well together to reduce the possibility of becoming stuck in local minima.

$$Y_j = \begin{cases} Y_j^{hippoS} G_j^{hippoS} > G_j \\ Y_j G_j^{hippoS} \geq G_j \end{cases} \quad (39)$$

When a hippopotamus confronts a group of predators or is unable to fend off the predator with its protective behavior, it will exhibit other behaviors. The hippopotamus attempts to leave the environment in this scenario. Spotted lions as well as hyenas tend not to venture into lakes or ponds, so when a hippopotamus senses danger,

it usually attempts to flee to the closest one. Using this tactic, the hippopotamus is able to locate a secure spot that is near to its present location. By simulating this behavior in third phase of the HO, this behavior may be better utilized for local search purposes. A random place is created close to the hippocampal locations in order to replicate this behavior. The hippopotamuses' behavior is represented using Eqs. (40–43). The hippopotamus has altered its location since it has discovered a safer spot close to its existing position when the newly generated location increases the cost function value. U stands for the MaxIter and u for the present iteration.

$$LB_k^{Local} = \frac{LB_k}{u}, UB_k^{Local} = \frac{UB_k}{u}, u = 1, 2, \dots, U \quad (40)$$

$$Y_j^{hippo\epsilon} : y_{jk}^{hippo\epsilon} = y_{jk} + s_{10} \dots (LB_k^{Local} + t_1 \dots (UB_k^{Local} - LB_k^{Local})) j = 1, 2, \dots, O, k = 1, 2, \dots, n \quad (41)$$

The location related to the hippos was searched to determine the nearest safe spot, and this is represented by $Y_j^{hippo\epsilon}$ in Eq. (41). Three situations t are represented by the random vector or integer t_1 , which is chosen at random using Eq. (42). The cases under consideration (t) result in a local search that is more appropriate, or, to put it another way, the suggested algorithm has a greater exploitation quality.

$$t = \begin{cases} 2 \times \vec{s}_{11} - 1 \\ s_{12} \\ s_{13} \end{cases} \quad (42)$$

While s_{10} (Eq. 41) and s_{13} indicate random integers created within the range of 0 and 1, $\vec{s}_{11}$ in Eq. (42) denotes a random vector between 0 and 1. Furthermore, s_{12} represents a random integer with a normal distribution.

$$Y_j = \begin{cases} Y_j^{hippo\epsilon} G_j^{hippo\epsilon} G_j \\ Y_j G_j^{hippo\epsilon} \geq G_j \end{cases} \quad (43)$$

The entire population individuals are updated on the basis of Phases 1 through 3 following the completion of every iteration of the HO algorithm. This procedure of updating the population in accordance with Eqs. (27–43) continues till the last iteration. As the algorithm runs, the optimal solution is continuously monitored and recorded. The optimal candidate—known as the dominant hippopotamus solution—is revealed as the final answer to the problem once the full procedure has run. Algorithm 1 displays the procedural specifics of the HO as pseudocode.

Algorithm 1: HO*Start**Describe an optimization problem [optimal coverage of the proposed WSN coverage model]**Place the hippopotamus count O and maximum iterations count U [sensor nodes count, monitoring area of the network parameters, sensing radius, and initial energy of the proposed WSN coverage model]*

$$Y_j: y_{jk} = LB_k + s \cdot (UB_k - LB_k), j = 1, 2, \dots, O, k = 1, 2, \dots, n$$

*For u = 1: U**Update the position of the dominant hippopotamus on the basis of the fitness function value criterion [optimal coverage of the proposed WSN coverage model]**Perform exploration phase by updating the position of the hippopotamus in the pond or river**For j = 1: O/2*

$$Y_j^{N \text{ Hippo}}: y_{jk}^{N \text{ Hippo}} = y_{jk} + z_1 \cdot (E \text{ Hippo} - J_1 y_{jk}) \text{ for } j = 1, 2, \dots, \left\lfloor \frac{O}{2} \right\rfloor \text{ and } k = 1, 2, \dots, n$$

$$Y_j^{GC \text{ Hippo}}: y_{jk}^{GC \text{ Hippo}} = \begin{cases} y_{jk} + i_1 \cdot (E \text{ Hippo} - J_2 N H_j) & U > 0.6 \\ \varepsilon & \text{otherwise} \end{cases}$$

$$Y_j = \begin{cases} Y_j^{N \text{ Hippo}} & G_j^{N \text{ Hippo}} < G_j \\ Y_j & \text{otherwise} \end{cases}$$

$$Y_j = \begin{cases} Y_j^{GC \text{ Hippo}} & G_j^{GC \text{ Hippo}} < G_j \\ Y_j & \text{otherwise} \end{cases}$$

*End for**Perform exploration phase by performing hippopotamus defense against the predators**For j = 1 + O/2: O*

$$\text{predator}: \text{predator}_k = LB_k + \tilde{s}_8 \cdot (UB_k - LB_k), k = 1, 2, \dots, n$$

$$Y_j^{\text{hippo } S}: y_{jk}^{\text{hippo } S}$$

$$= \begin{cases} \overline{SM} \oplus \text{predator}_k + \left(\frac{g}{(\sigma e \times \cos(2\pi h))} \right) \cdot \left(\frac{1}{\tilde{E}} \right) & G_{\text{predator}_k} < G_j \\ \overline{SM} \oplus \text{predator}_k + \left(\frac{g}{(\sigma e \times \cos(2\pi h))} \right) \cdot \left(\frac{1}{2 \times \tilde{E} + \tilde{s}_9} \right) & G_{\text{predator}_k} \geq G_j \end{cases} \text{ for } j$$

$$= \left\lfloor \frac{O}{2} \right\rfloor + 1, \left\lfloor \frac{O}{2} \right\rfloor + 2, \dots, O \text{ and } k = 1, 2, \dots, n$$

$$Y_j = \begin{cases} Y_j^{\text{hippo } S} & G_j^{\text{hippo } S} < G_j \\ Y_j & G_j^{\text{hippo } S} \geq G_j \end{cases}$$

*End for**Perform the exploitation phase with consideration of the hippopotamus escaping from the predator*

$$LB_k^{\text{Local}} = \frac{LB_k}{u}, UB_k^{\text{Local}} = \frac{UB_k}{u}, u = 1, 2, \dots, U$$

For j = 1: O

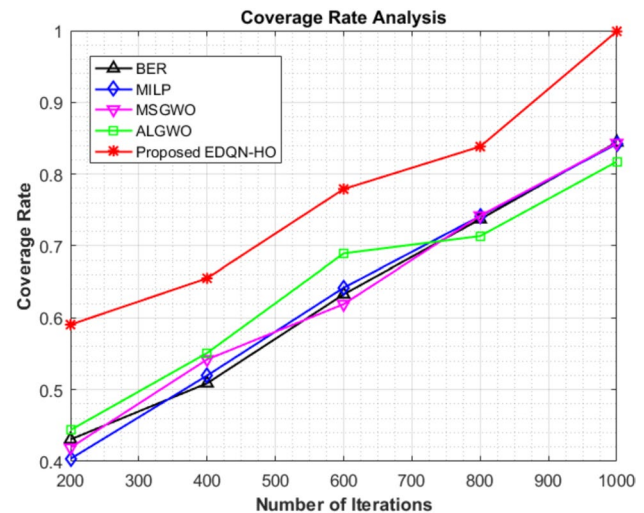
$$Y_j^{\text{hippo } \varepsilon}: y_{jk}^{\text{hippo } \varepsilon} = y_{jk} + s_{10} \cdot (LB_k^{\text{Local}} + t_1 \cdot (UB_k^{\text{Local}} - LB_k^{\text{Local}})) \text{ } j = 1, 2, \dots, O, k = 1, 2, \dots, n$$

$$Y_j = \begin{cases} Y_j^{\text{hippo } \varepsilon} & G_j^{\text{hippo } \varepsilon} < G_j \\ Y_j & G_j^{\text{hippo } \varepsilon} \geq G_j \end{cases}$$

*End for**Save the optimal candidate solution attained so far [best optimal coverage]**End for**Output the best solution associate with the fitness function attained by HO [optimal coverage of the proposed WSN coverage model]**End HO**Stop*

Table 1 Coverage rate analysis

Methods	Iterations				
	200	400	600	800	1000
BER [17]	0.4306	0.5089	0.6328	0.7372	0.8444
MILP [21]	0.4036	0.5197	0.6420	0.7415	0.8421
MSGWO [24]	0.4192	0.5421	0.6192	0.7422	0.8433
ALGWO [25]	0.4438	0.5511	0.6896	0.7137	0.8174
Proposed EDQN-HO	0.5907	0.6552	0.7792	0.8384	0.9990

Fig. 5 Coverage rate analysis


4 Results and Analysis

4.1 Experimental Setup

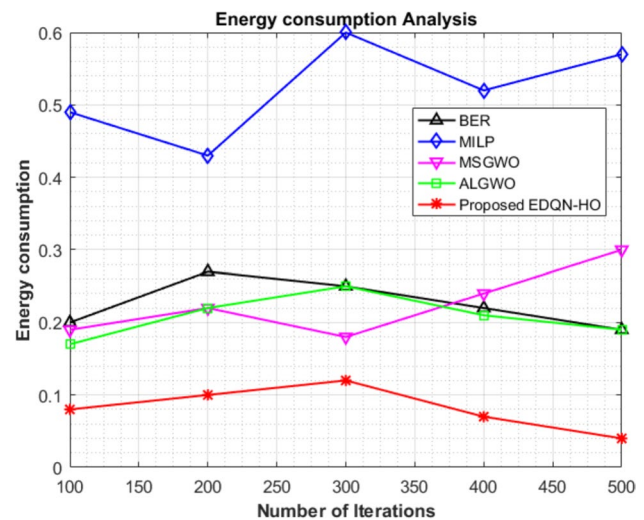
The proposed EDQN-HO algorithm for the coverage optimization in WSNs was implemented in MATLAB and the findings were analyzed. The population size and the iteration count were placed to be 10 and 1000. The total number of nodes considered is assumed to be 500. The proposed model was compared with distinct conventional models like BER [17], MILP [21], MSGWO [24], and ALGWO [25] with respect to analysis such as coverage rate, energy consumption, computing time, positioning error, network lifetime, and average moving distance to reveal the supremacy of the developed coverage optimization in WSNs model.

4.2 Coverage Rate Analysis

Table 1 and Fig. 5 affords a contrast of coverage rate across numerous strategies at distinctive iterations, highlighting the overall performance of the proposed EDQN-HO model within the context of coverage optimization in WSNs. As the variety of iterations increases, the coverage rate commonly decreases for all techniques, indicating advanced reliability in statistics transmission. Notably, the proposed EDQN-HO achieves the lowest coverage rate across all iteration counts, especially excelling with a super reduction to 0.9990 at 1000 iterations. This shows that the EDQN-HO technique drastically complements coverage performance, ensuring better facts integrity and efficiency. In comparison, traditional strategies along with MILP and MSGWO exhibit quite better coverage rate values, emphasizing the benefits of the proposed EDQN-HO method in optimizing coverage in WSNs. Overall, the effects underscore the importance of superior algorithms in accomplishing advanced coverage and reliability in sensor network deployments. The proposed EDQN-HO for the WSN coverage

Table 2 Energy consumption analysis

Methods	Number of nodes				
	100	200	300	400	500
BER [17]	0.2	0.27	0.25	0.22	0.19
MILP [21]	0.49	0.43	0.6	0.52	0.57
MSGWO [24]	0.19	0.22	0.18	0.24	0.3
ALGWO [25]	0.17	0.22	0.25	0.21	0.19
Proposed EDQN-HO	0.08	0.1	0.12	0.07	0.04

Fig. 6 Energy consumption analysis

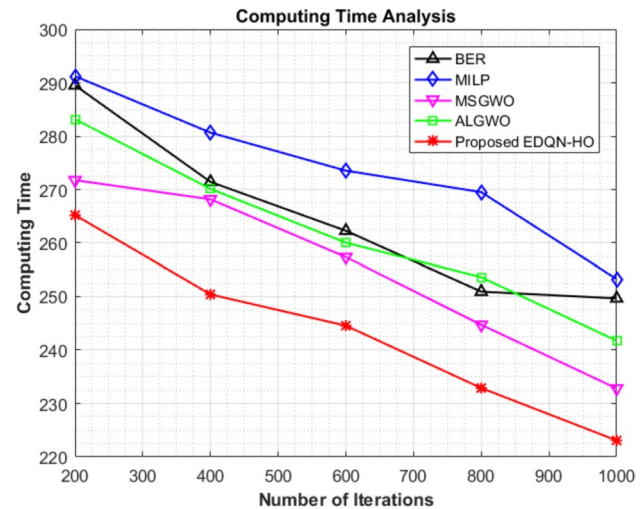
model in terms of coverage rate is 18.31%, 18.63%, 18.46%, and 22.22% better than BER, MILP, MSGWO, and ALGWO respectively.

4.3 Energy Consumption Analysis

Table 2 and Fig. 6 offer energy consumption throughout numerous techniques concerning specific numbers of nodes in WSNs, relative to coverage optimization. As the wide variety of nodes increases, the energy consumption varies many of the special strategies, reflecting their performance in coping with energy and maintaining facts integrity. The proposed EDQN-HO shows an enormous advantage, demonstrating the lowest energy consumption throughout all node counts, specially reaching an excellent 0.04 at 500 nodes. This suggests that the EDQN-HO method no longer best complements coverage however additionally optimizes energy consumption effectively, main to decreased records transmission errors. In comparison, traditional methods like MILP and MSGWO exhibit better energy consumption values, indicating less efficient energy usage and greater susceptibility to transmission errors as node count increases. The outcomes suggest that the proposed EDQN-HO version can keep a higher degree of overall performance with fewer retransmissions, in the long run keeping energy consumption. This is vital for WSNs, where energy consumption directly impacts reliability. By minimizing energy consumption while coping with an increasing number of nodes, the EDQN-HO approach underlines the significance of intelligent algorithms in sensor networks, making it a most effective preference for optimizing WSN overall performance. The proposed EDQN-HO for the WSN coverage model with respect to energy consumption is 78.95%, 92.98%, 86.67%, and 78.95% advanced than BER, MILP, MSGWO, and ALGWO respectively.

Table 3 Computing time analysis

Methods	Iterations				
	200	400	600	800	1000
BER [17]	289.4996	271.4363	262.2828	250.9370	249.6768
MILP [21]	291.2138	280.6576	273.5338	269.5167	253.2707
MSGWO [24]	271.7538	268.1806	257.3758	244.7328	232.8174
ALGWO [25]	283.1568	270.1280	260.0654	253.6211	241.7490
Proposed EDQN-HO	265.2170	250.4234	244.5706	232.8963	223.0806

Fig. 7 Computing time analysis


4.4 Computing Time Analysis

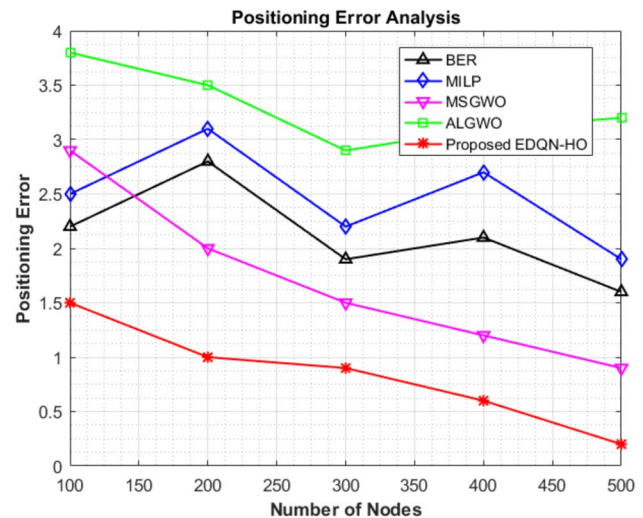
Table 3 and Fig. 7 illustrate the computing time associated with various techniques throughout specific iterations, presenting insights into their efficiency in the context of coverage optimization in WSNs. As the quantity of iterations increases, computing time commonly fluctuates in many of the unique methods, reflecting their computational complexity and overall performance. The proposed EDQN-HO method consistently demonstrates the lower computing time throughout all iterations. This trend shows that the efficiency of EDQN-HO is somehow enhanced. This is more so the case in WSNs, as it is mainly about how well decision making within real time should be processed. And even though MILP can produce some reasonable solutions, it can also mean considerable degree of computation, which probably limits its practical use in dynamic WSN applications. Likewise, both MSGWO and ALGWO show off small computing periods in their runtime, showing a trade-off between performance and efficiency. But their level of performance is not within the level of the proposed EDQN-HO. In all instances, the consequences emphasize the importance of selecting algorithms that optimize coverage and computing available time rendering it unnecessary to compromise the general performance of the WSNs and thereby lengthening their life span and improving their reliability in varied fields of application. The proposed EDQN-HO for the WSN coverage model in terms of computing time is 10.65%, 11.92%, 4.18%, and 7.72% higher than BER, MILP, MSGWO, and ALGWO, respectively.

4.5 Positioning Error Analysis

Table 4 and Fig. 8 present the positioning errors obtained when different approaches are employed in the coverage optimization of WSNs with varying numbers of nodes and complexity. The positioning errors are crucial in a WSN since they affect the quality of data acquisition as well as the operation of the network. Most notably, the

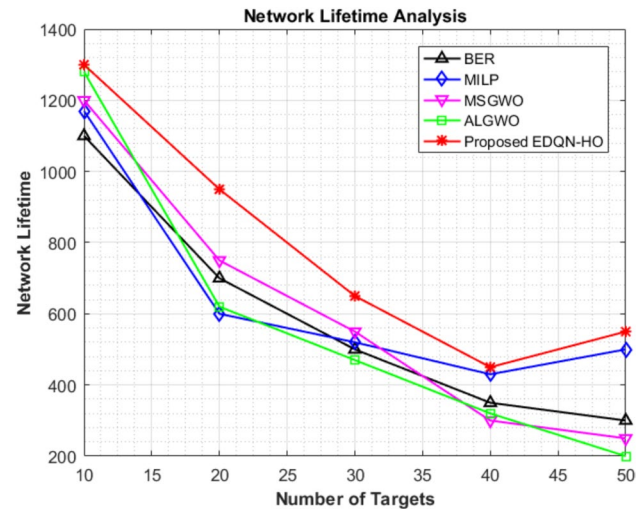
Table 4 Positioning error analysis

Methods	Number of nodes				
	100	200	300	400	500
BER [17]	2.2	2.8	1.9	2.1	1.6
MILP [21]	2.5	3.1	2.2	2.7	1.9
MSGWO [24]	2.9	2	1.5	1.2	0.9
ALGWO [25]	3.8	3.5	2.9	3.1	3.2
Proposed EDQN-HO	1.5	1	0.9	0.6	0.2

Fig. 8 Positioning error analysis**Table 5** Network lifetime analysis

Methods	Number of targets				
	10	20	30	40	50
BER [17]	1100	700	500	350	300
MILP [21]	1170	600	520	430	500
MSGWO [24]	1200	750	550	300	250
ALGWO [25]	1280	620	470	320	200
Proposed EDQN-HO	1300	950	650	450	550

proposed EDQN-HO method performed well with all node counts by demonstrating a consistent low error count. This confirms that the EDQN-HO method not only enhances coverage efficiency by taking into account the node positions but also remarkably enhances the precision of positioning. Positioning errors are at a minimum and these are perceived to limit the collection of data in the WSN depending on the amount of coverage needed in certain applications. This is in sharp contrast to approaches like ALGWO where positioning errors are significantly high. This variation is probably attributed to the increasing node numbers such that the performance of ALGWO technique in reducing positioning error reduces notable coverage enhancement, data loss. Likewise, positioning error values are also exhibited by MILP and MSGWO that show degrading performance. The resultant pertains to attempts which focus on improvement of existing methods demonstrates accomplishment by making coverage optimization in WSN even better by paying attention to positioning error reduction. The proposed EDQN-HO for the WSN coverage model with respect to positioning error is 87.5%, 89.47%, 77.78%, and 93.75% more than BER, MILP, MSGWO, and ALGWO respectively.

Fig. 9 Network lifetime analysis

4.6 Network Lifetime Analysis

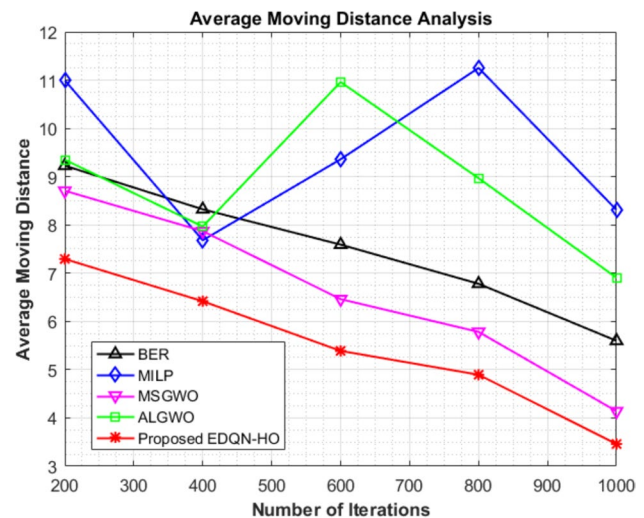
Table 5 and Fig. 9 summarize the trend of networks lifetime for various methods based on numbers of target in WSNs. This observation is important to understand the effect of coverage optimization on network lifetime, which is an important metric of network lifespan during WSN applications because both energy efficiency and data reliability are found to be paramount. The suggested EDQN-HO technique reveals a steady growth of network lifetime. On the other hand, methods such as ALGWO and MSGWO record greater network longevity values, of course with fewer targets, which may imply problems in data integrity when the system is subjected to a more intensive load. MILP also reports noticeable improvements in the network lifespan when its number of targets is increased, though with a rather slight improvement at higher counts. Though the proposed EDQN-HO method has higher initial values of lifespan, it performs better overall in handling the coverage for multiple targets by gaining more efficiency with respect to energy and also extended network lifespan. With increased growth in the network, EDQN-HO holds lower lifespans; this means EDQN-HO preserves better sustainability of an operational efficiency aspect in WSNs which in turn therefore means they can be used for longer periods without losing their dependability in various applications. The proposed EDQN-HO for the WSN coverage model in terms of network lifetime is 83.33%, 9.09%, 54.55%, and 63.64% greater than BER, MILP, MSGWO, and ALGWO respectively.

4.7 Average Moving Distance Analysis

Table 6 and Fig. 10 represent average distance moved through different methods of various iterations, which can effectively be shown as input for optimizing coverage in WSNs. Average distance moved is an important parameter that shows how efficiently nodes can move and rearrange to meet the requirements of the network and directly correlates to the quality of coverage and energy efficiency. Among all of them, the proposed EDQN-HO

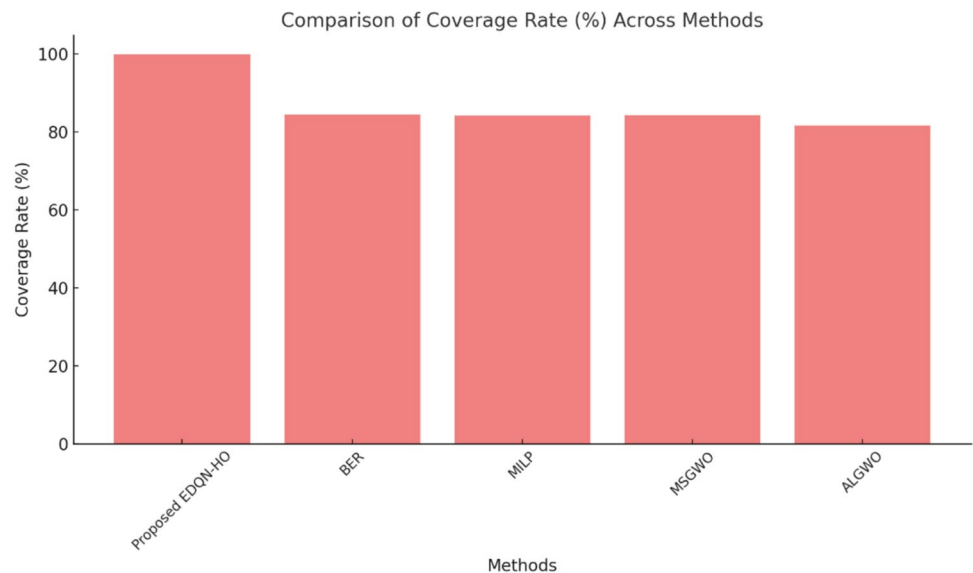
Table 6 Average moving distance analysis

Methods	Iterations				
	200	400	600	800	1000
BER [17]	9.2271	8.3234	7.5918	6.7804	5.5988
MILP [21]	10.9958	7.6821	9.3612	11.2535	8.3102
MSGWO [24]	8.7083	7.8718	6.4619	5.7833	4.1400
ALGWO [25]	9.3507	7.9669	10.9614	8.9682	6.9053
Proposed EDQN-HO	7.2943	6.4186	5.3896	4.8956	3.4600

Fig. 10 Average moving distance analysis

method always performs with the minimum possible distances at all iterations. Because this drop is indicative of the fact that EDQN-HO algorithm improves not only coverage but also decreases the error as nodes are moving and changing their positions. However, other conventional methods display inconsistency in their strategy for movement that might result in more energy consumption and shorter lifetime for the network. This means that MILP may have difficulties in sustaining optimal node locations when network conditions vary. Although the overall performance of other methods is good, it still does not match up to the effectiveness of the proposed EDQN-HO method. This method truly shines by maintaining minimal distances across iterations, showcasing superior efficiency in node placement which results in lower energy usage and higher coverage reliability. Hence, for applications requiring strong performance and durability in dynamic environments, the EDQN-HO method proves to be an excellent choice for enhancing WSN deployments effectively. The proposed EDQN-HO for the WSN coverage model with respect to average moving distance is 38.20%, 58.36%, 16.43%, and 49.89% superior to BER, MILP, MSGWO, and ALGWO respectively.

Figure 11 illustrates the comparison of **coverage rate (%)** across various methods, including the proposed EDQN-HO algorithm, BER, MILP, MSGWO, and ALGWO. The coverage rate is a critical metric for WSNs as it measures the proportion of the monitored area effectively covered by the sensor nodes.

Fig. 11 Coverage rate analysis

The proposed EDQN-HO algorithm outperforms all other methods, achieving a coverage rate of **99.9%**, significantly higher than the closest competitors (BER: 84.44%, MILP: 84.21%, MSGWO: 84.33%, ALGWO: 81.74%). This improvement demonstrates the robustness of the EDQN-HO in optimizing node placement and activation, leading to near-complete coverage of the monitored area. The substantial margin of superiority (over 18% improvement compared to conventional methods) highlights the advanced capability of the EDQN-HO algorithm to adapt dynamically to network conditions, ensuring minimal blind spots and optimal resource utilization. These attributes make the proposed method highly effective for real-world WSN applications requiring high coverage reliability.

5 Conclusion

This research proposed a novel intelligent deep learning-based optimization strategy for WSN coverage optimization. In this case, the unique EDQN method achieved the best network coverage of WSN, and the fitness function was attained by fine-tuning the parameters of DQN using the nature-inspired optimization algorithm HO. The established approach modified the node location to represent the spatial features of the network to maximize coverage. The enhanced coverage not only reduced latency but also boosted throughput and extended the life of the network. The results of several network coverage optimization experiments were calculated, and by varying the parameters, the effect of the EDQN-HO on network coverage optimization was also ascertained. The simulated results from the network coverage optimization experiments showed that the proposed EDQN-HO might be used successfully in a range of scenarios. The proposed EDQN-HO for the WSN coverage model in terms of coverage rate was 18.31%, 18.63%, 18.46%, and 22.22% better than BER, MILP, MSGWO, and ALGWO respectively. Similarly, the proposed EDQN-HO for the WSN coverage model with respect to energy consumption was 78.95%, 92.98%, 86.67%, and 78.95% advanced than BER, MILP, MSGWO, and ALGWO respectively.

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Declarations

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References

- Chand, S., Kumar, B.: Genetic algorithm-based meta-heuristic for target coverage problem. *IET Wireless Sensor Syst.* **8**(4), 170–175 (2018)
- Farsi, M., Elhosseini, M. A., Badawy, M., Arafat, H., ZainEldin, H.: Deployment techniques in wireless sensor networks, coverage and connectivity: A survey. *IEEE Access* (2019)
- Qin, D., Ma, J., Zhang, Y., Feng, P., Ji, P., Berhane, T.M.: Study on connected target coverage algorithm for wireless sensor network. *IEEE Access.* **6**, 69415–69425 (2018)
- Luo, C., Hong, Y., Li, D., Wang, Y., Chen, W., Hu, Q.: Maximizing network lifetime using coverage sets scheduling in wireless sensor networks. *Ad Hoc Networks.* 102037 (2020)
- Biswas, S., Das, R., Chatterjee, P.: Energy-Efficient Connected Target Coverage in Multi-hop Wireless Sensor Networks. *Indus Interactive Innovations in Science, Engineering and Technology Springer.* pp. 411–421 (2018)
- Gupta, G.P., Jha, S.: Biogeography-based optimization scheme for solving the coverage and connected node placement problem for wireless sensor networks. *Wireless Netw.* **25**(6), 3167–3177 (2019)
- Elhoseny, M., Tharwat, A., Yuan, X., Hassanien, A.E.: Optimizing K-coverage of mobile WSNs. *Expert Syst. Appl.* **92**, 142–153 (2018)
- Etancelin, J.-M., Fabbri, A., Guinand, F., Rosalie, M.: DACYCLEM: A decentralized algorithm for maximizing coverage and lifetime in a mobile wireless sensor network. *Ad Hoc Netw.* **87**, 174–187 (2019)
- Harizan, S., Kuila, P.: Coverage and connectivity aware energy efficient scheduling in target based wireless sensor networks: an improved genetic algorithm-based approach. *Wireless Netw.* **25**(4), 1995–2011 (2019)
- Idrees, A.K., Deschinkel, K., Salomon, M., Couturier, R.: Multiround distributed lifetime coverage optimization protocol in wireless sensor networks. *J. Supercomput.* **74**(5), 1949–1972 (2018)
- Muhammad, Z., Roy, A., Ahn, C. W., Sachan, R., Saxena, N.: A novel random scheduling algorithm based on subregions coverage for SET K-Cover problem in wireless sensor networks. *KSII Trans. Internet Inform. Syst.* **12**(6) (2018)
- Katti, A.: Target coverage in random wireless sensor networks using cover sets. *J King Saud University Comput. Inform. Sci.* (2019)
- Anitha Jebamani Soundararaj: Godfrey Winster Sathianesan, "Task offloading scheme in Mobile Augmented Reality using hybrid Monte Carlo tree search (HMCTS)". *Alex. Eng. J.* **108**, 611–625 (2024)
- Samuel Manoharan, J.: A metaheuristic approach towards enhancement of network lifetime in wireless sensor networks. *KSII Trans. Internet Inf. Syst.* **17**(4), 1276–1295 (2023)
- Manoharan, J.S.: Double attribute-based node deployment in wireless sensor networks using novel weight-based clustering approach. *Sadhana* **47**(3), 1–11 (2022)
- Keshmiri, H., Bakhshi, H.: A new 2-phase optimization-based guaranteed connected target coverage for wireless sensor networks. In *IEEE Sensors J.* **20**(13) 7472–7486 (2020)
- Kumar, S., Jain, S., Sharma, A.: A BER-conscious synthesis of switched-beam antenna to maximize reliable coverage in wsn applications. In *IEEE Sensors J.* **22**(4), 3785–3795 (2022)
- Jia, R., Zhang, H.: Wireless Sensor Network (WSN) model targeting energy efficient wireless sensor networks node coverage. *IEEE Access* **12**, 27596–27610 (2024)
- Hao, Z., Qu, N., Dang, X., Hou, J.: RSS-based coverage deployment method under probability model in 3D-WSN. *IEEE Access* **7**, 183091–183104 (2019)
- Kou, G., Wei, G.: Improved sparrow search algorithm optimized DV-Hop for wireless sensor network coverage. *IEEE Access* **11**, 62348–62357 (2023)
- Kumari, S., Srirangarajan, S.: Node placement and path planning for improved area coverage in mixed wireless sensor networks. *IEEE Robot. Autom. Lett.* **9**(8), 6800–6807 (2024)
- Ma, N., Wang, S., Hao, S.: Enhancing Reptile search algorithm with shifted distribution estimation strategy for coverage optimization in wireless sensor networks. *Heliyon* **10**(15), (2024)
- Zulfiqar, R., Javid, T., Ali, Z. A., Uddin, V.: Novel metaheuristic routing algorithm with optimized energy and enhanced coverage for WSNs. *Ad Hoc Networks*, **144** (2023)
- Wang, Z., Huang, L., Yang, S., Luo, X., He, D., Chan, S.: Multi-strategy enhanced grey wolf algorithm for obstacle-aware WSNs coverage optimization. *Ad Hoc Networks*, **152** (2024)
- Yu, X., Duan, Y., Cai, Z., Luo, W.: An adaptive learning grey wolf optimizer for coverage optimization in WSNs. *Expert Syst. Appl.* **238** (2024)
- Wajgi, D., Thakur, N.V.: Load balancing based approach to improve lifetime of wireless sensor network. *Int J Wireless Mobile Netw* **4**(4), 155 (2012)
- Wajgi, D.W., Tembhurne, J.V.: Localization in wireless sensor networks and wireless multimedia sensor networks using clustering techniques. *Multimedia Tools Appl* **83**(3), 6829–6879 (2024)

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